



SZABIST
UNIVERSITY
Islamabad Campus

Shaheed Zulfikar Ali Bhutto Institute of Science & Technology

COMPUTER SCIENCE DEPARTMENT

Total Marks: 04

Obtained Marks: _____

Parallel and Distributed Computing

Assignment # 04

Submitted To: Rana Faisal Hayat

Student Name: Ubaid Bin Waris

Reg Number: 2212416

COMPUTER SCIENCE DEPARTMENT

Question no. 1

Why do we use HDFS for applications having large datasets rather than applications having small files

HDFS (Hadoop file system) is specially design to store and process very large amount of data. It work efficiently when files are larger because it divides files into large blocks (typically 128 mb or more) and distributes them across multiple machines in a hadoop cluster.

Problems with many small files in HDFS

Metadata overload

HDFS store metadata of every file and block in the Name Node's memory.

Thousands or Million of small files consumes a large amount of RAM.

Inefficient Storage

Default HDFS block size typically 128 MB or larger.

A 1 kb file still requires metadata manghment similar to long file.

Poor performance

Accessing many small files increasing communication with NameNode

This create bottlenecks and reduce overall system performance.

COMPUTER SCIENCE DEPARTMENT

Advantages of large files in HDFS

- Better utilization of large block size
- Fewer metadata entries
- Higher read/write throughput
- More efficient MapReduce Processing

HDFS is design for storing and processing large datasets (GBs/TBs) rather than million of small files.

Question no. 2

How do you define rack awareness in Hadoop? How does it help in fault tolerance?

Rack awareness is a hadoop feature that understand the network topology of a cluster and know which DataNodes belongs to which rack.

A Rack is a collection of servers connected through the same network switch.

How it works

When HDFS replicates a block (default replication factor=3)

1st replica → stored on local Node

2nd replica → store on a node in different rack

3rd replica → store on another node within the second rack.

COMPUTER SCIENCE DEPARTMENT

Benefits of Fault Tolerance

- Reck failure protection
- Data Availability
- Improved Reliability
- Optimize network traffic

Question no. 3

Simple Request-Response Client-Server Code
(Python sockets)

Server Code

```
# Import python's sockets module
import socket
```

```
# create TCP object, socket
server-socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
```

```
# bind object to localhost and port 5000
server-socket.bind(("localhost", 5000))
```

```
# Start listening for incoming connection
server-socket.listen(1)
```

```
Print("Server is waiting for connection...")
```

```
# Accept a client connection
client-socket, client-address = server-socket.accept()
```


COMPUTER SCIENCE DEPARTMENT

```
Print ("Connected by: ", client_address)

# Recive up to 1024 byte from the client
request = client_socket.recv(1024).decode()

Print ("Client says:", request)

# Prepare response message
response = "Hello client, request recived"

# Send response back to client
client_socket.send(response.encode())

# Close client connection
client_socket.close()

# Close server connection
server_socket.close()
```

Client code

```
# Import python socket module
import socket

# Create TCP object, socket
client_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

# Connect to server running on localhost port 5000
client_socket.connect(("localhost", 5000))

message = "Hello server"

# send msg to server
client_socket.send(message.encode())

# Recive response from server
response = client_socket.recv(1024).decode()

Print ("Server reply:", response)

# Close connection
client_socket.close()
```